

Table 61. IWDG min/max timeout period at 32 kHz (LSI)

For the values in this table, the exact timings still depend on the phasing of the APB interface clock versus the LSI clock, so that there is always a full RC period of uncertainty.

Prescaler divider	PR[2:0] bits	Min timeout RL[11:0] = 0x000	Max timeout RL[11:0] = 0xFFFF	Unit
/4	0	0.125	512	ms
/8	1	0.250	1024	
/16	2	0.500	2048	
/32	3	1.0	4096	
/64	4	2.0	8192	
/128	5	4.0	16384	
/256	6 or 7	8.0	32768	

Table 62. WWDG min/max timeout value at 144 MHz (PCLK)

Prescaler	WDGTB	Min timeout values	Max timeout value	Unit
1	0	0.0284	1.820	ms
2	1	0.0569	3.641	
4	2	0.1138	7.282	
8	3	0.2276	14.564	
16	4	0.4551	29.127	
32	5	0.9102	58.254	
64	6	1.820	116.508	
128	7	3.641	233.017	

5.3.23 I3C interface characteristics

The I3C interface meets the timing requirements of the MIPI® I3C specification v1.1.

The I3C peripheral supports:

- I3C SDR-only as controller
- I3C SDR-only as target
- I3C SCL bus clock frequency up to 12.5 MHz

The parameters given in Table 63 below are obtained with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- I/O Compensation cell activated
- Voltage scaling range 1

The I3C timings are in line with the MIPI specification, except for the ones given in Table 63, I3C open-drain measured timing. For t_{SU_OD} , this can be mitigated by increasing the corresponding SCL low duration in the I3C_TIMINGR0 register. For further details refer to AN5879.

Table 63. Open drain timing measurements

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions	Min	Unit
t_{SU_OD}	SDA data setup time in open drain mode	Controller $2.7\text{ V} \leq V_{DDIOX} \leq 3.6\text{ V}$	22 ⁽¹⁾	ns

1. The minimum SDA data setup time during open-drain-mode is 3 ns, as specified in the MIPI Alliance specification for I3C.